

Weather Trend Forecasting using Machine Learning

PM Accelerator Technical Assessment

Candidate: Rramandeip Singh

Dataset: Global Weather Repository (Kaggle)

Introduction

This project presents a comprehensive analysis of the Global Weather Repository dataset to understand weather patterns and develop machine learning models for temperature forecasting. The dataset contains more than 152,000 weather observations collected from cities worldwide and includes 41 weather-related features, such as temperature, humidity, atmospheric pressure, wind speed, precipitation, visibility, UV index, and air quality indicators. The project begins with data preprocessing, including data cleaning, duplicate removal, handling missing values, and converting date fields into an appropriate format to ensure data quality before analysis.

Following preprocessing, extensive Exploratory Data Analysis (EDA) was performed using descriptive statistics and visualisations to identify trends, distributions, and relationships among weather variables. Three forecasting models—Linear Regression, Random Forest, and XGBoost—were developed and evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R^2 Score. In addition, feature importance analysis was conducted to determine the variables that contribute most to temperature prediction, while climate trend analysis, air quality analysis, and geographical analysis provided further insights into environmental conditions across different countries. Overall, the project demonstrates how machine learning can effectively analyse complex weather data and improve forecasting accuracy using advanced ensemble learning techniques.

PM Accelerator Mission

PM Accelerator is a career-focused program designed to help aspiring AI engineers, product managers, designers, and data professionals gain practical industry experience. The program emphasizes learning through real-world projects, allowing participants to apply artificial intelligence, machine learning, and data analysis concepts to solve business problems. In addition to technical training, participants receive mentorship from experienced professionals, collaborate with multidisciplinary teams, and develop industry-ready portfolios. The program also focuses on strengthening problem-solving, communication, and project management skills, preparing learners for careers in AI and technology. Overall, PM Accelerator bridges the gap between academic knowledge and

industry expectations by providing hands-on experience in developing practical AI solutions.

Figure 1. Dataset Overview

The Global Weather Repository dataset contains weather observations collected from cities and regions worldwide, making it suitable for global climate and forecasting studies. It includes more than 152,000 records and 41 weather-related features, covering atmospheric, environmental, geographical, and air quality information. Key variables include temperature, humidity, atmospheric pressure, wind speed, wind direction, precipitation, cloud cover, visibility, UV index, latitude, longitude, sunrise and sunset times, and timestamps. The dataset also contains air quality indicators such as PM2.5, PM10, carbon monoxide, nitrogen dioxide, ozone, and sulphur dioxide, providing a comprehensive foundation for weather analysis, visualisation, and machine-learning-based temperature prediction.

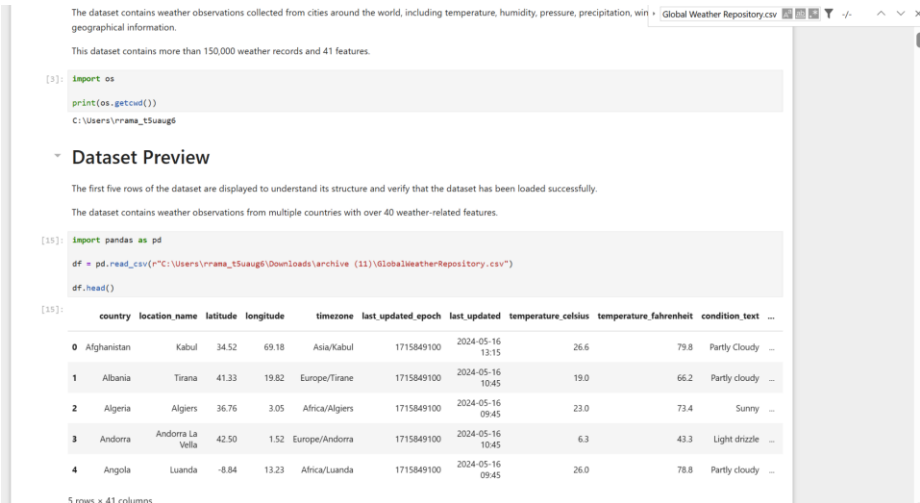
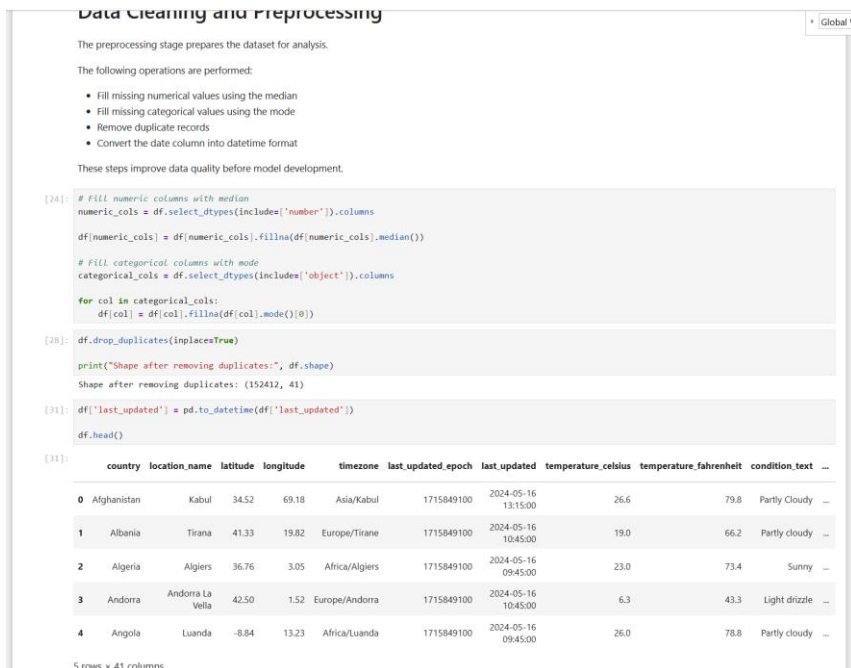


Figure 2. Data Cleaning and Preprocessing

Before performing analysis, the dataset underwent several data preprocessing and quality assurance steps to improve its reliability. Missing values were identified and handled appropriately by filling numerical columns with the median and categorical columns with the mode wherever necessary. Duplicate records were checked and removed to eliminate redundant observations and ensure data consistency. The last_updated column was converted into the datetime format, enabling accurate time-series analysis and forecasting. In addition, numerical features were verified and prepared for machine learning models by ensuring they had appropriate data types. These preprocessing steps improved data quality, reduced potential errors, and created a clean, consistent dataset suitable for visualisation, statistical analysis, and predictive modelling.



Exploratory Data Analysis

Exploratory Data Analysis (EDA) was conducted to better understand the structure, trends, and relationships within the weather dataset before developing forecasting models. Several visualizations were created, including temperature trends over time, temperature and precipitation distributions, correlation heatmaps, boxplots for identifying potential outliers, and scatter plots showing the relationship between humidity and temperature. These analyses provided valuable insights into the distribution of weather variables and helped identify patterns that could influence temperature prediction. The EDA process also highlighted correlations between environmental factors, making it easier to select meaningful features for machine learning models and supporting more accurate and reliable weather forecasting.

Figure 3. Forecasting Models

To evaluate forecasting performance, three different machine learning models were developed and compared. Linear Regression was first implemented as a baseline model using the date feature to predict temperature over time. Since weather patterns are influenced by multiple environmental factors, more advanced models were also developed. Random Forest and XGBoost were trained using several weather-related features, including humidity, pressure, wind speed, precipitation, cloud cover, UV index, and visibility. The performance of each model was evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R^2 Score. Finally, a simple ensemble model combined predictions from all three models to assess whether combining algorithms could improve overall forecasting accuracy and model reliability.

Forecasting using Linear Regression

The first forecasting model uses Linear Regression.

The **last_updated** date is converted into an ordinal value so that it can be used as a numerical feature.

Model evaluation metrics include:

- MAE
- RMSE
- R² Score

```
[109]: from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42
)
```

```
[110]: from sklearn.linear_model import LinearRegression
lr = LinearRegression()
lr.fit(X_train, y_train)
```

```
[111]: > LinearRegression
Parameters
```

```
[112]: pred_lr = lr.predict(X_test)
```

Linear Regression Results

The model performance is evaluated using standard regression metrics.

These metrics indicate how accurately the model predicts future temperature values.

```
[120]: from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score
import numpy as np

print("MAE :", mean_absolute_error(y_test, pred_lr))
print("RMSE:", np.sqrt(mean_squared_error(y_test, pred_lr)))
print("R^2 Score:", r2_score(y_test, pred_lr))

MAE : 7.339959241908088
RMSE: 9.328184582379234
R^2 Score: 0.028352697812728356
```

```
[121]: import matplotlib.pyplot as plt

plt.figure(figsize=(10,6))

plt.scatter(X_test, y_test, alpha=0.5, label="Actual")
plt.scatter(X_test, pred_lr, alpha=0.5, label="Predicted")

plt.xlabel("Date (Ordinal)")
plt.ylabel("Temperature (°C)")
plt.title("Linear Regression Forecast")
plt.legend()

plt.show()
```

Linear Regression Forecast

Figure 4. Advanced Analysis

In addition to forecasting, several advanced analyses were performed to gain deeper insights into the weather dataset. Feature importance analysis using the Random Forest model identified the most influential variables affecting temperature prediction, helping improve model interpretability. Isolation Forest was applied to detect unusual weather observations and identify potential anomalies within the dataset. An air quality analysis examined the relationship between PM2.5 levels and temperature to better understand environmental conditions. Climate trend analysis explored changes in average temperatures over time, while geographical analysis compared average temperatures across different countries, highlighting regional climate variations and providing a broader understanding of global weather patterns..

Forecasting using Random Forest

Random Forest is an ensemble learning algorithm capable of modeling complex nonlinear relationships.

Weather-related features used include:

- Humidity
- Pressure
- Wind Speed
- Cloud Cover
- UV Index
- Visibility
- Precipitation

The model is expected to outperform Linear Regression.

```
[136]: from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test = train_test_split(
    X, y,
    test_size=0.2,
    random_state=42
)

[137]: from sklearn.ensemble import RandomForestRegressor
rf = RandomForestRegressor(n_estimators=100, random_state=42)

[138]: rf.fit(X_train, y_train)

[139]: rf  # RandomForestRegressor
```

```
[140]: pred_rf = rf.predict(X_test)
```

Random Forest Performance

Random Forest predictions are evaluated using:

- Mean Absolute Error (MAE)
- Root Mean Squared Error (RMSE)
- R² Score

The results are compared with the Linear Regression model.

```
[141]: from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score
import numpy as np

print("Random Forest Results")
print("-----")
print("MAE : ", mean_absolute_error(y_test, pred_rf))
print("RMSE : ", np.sqrt(mean_squared_error(y_test, pred_rf)))
print("R² : ", r2_score(y_test, pred_rf))

Random Forest Results
-----
MAE : 0.287079108328778
RMSE : 0.358015892346609
R² : 0.7281547853413251
```

Feature Importance Analysis

Results

The forecasting models were evaluated to determine which algorithm produced the most accurate temperature predictions. Among the models tested, Random Forest delivered the best overall performance, achieving the lowest prediction errors and the highest R² Score. Compared with the baseline Linear Regression model, Random Forest was much more effective because it utilized multiple weather-related features rather than relying only on the date. XGBoost also performed well and produced competitive results, demonstrating its ability to model complex relationships within the dataset. Overall, the results show that ensemble-based machine learning models are better suited for weather forecasting, as they can capture nonlinear patterns and provide more accurate and reliable temperature predictions.

.Discussion

The results of this study demonstrate that weather forecasting can be significantly improved by using advanced machine learning models capable of capturing complex, non-linear relationships within the data. Unlike the baseline Linear Regression model, ensemble-based algorithms such as Random Forest and XGBoost were able to use multiple environmental variables simultaneously, resulting in more accurate temperature predictions. Feature importance analysis showed that variables including humidity, atmospheric pressure, UV index, precipitation, wind speed, and cloud cover play a major role in forecasting temperature. These findings highlight that weather prediction depends on the interaction of several environmental factors rather than a single variable, emphasizing the importance of using comprehensive datasets and robust machine learning techniques.

Executive Summary with Screenshots

This report presents a comprehensive weather forecasting project using the Global Weather Repository dataset and documents the complete data science workflow from data preparation to model evaluation. The project includes data cleaning, preprocessing, exploratory data analysis (EDA), and the development of multiple machine learning models, including Linear Regression, Random Forest, XGBoost, and an Ensemble model. Various visualizations and statistical analyses were performed to better understand weather patterns, feature relationships, and environmental trends. The report also includes feature importance analysis, anomaly detection, air quality analysis, climate trend analysis, and geographical comparisons. Screenshots of the key outputs are provided throughout the report to support the findings and clearly demonstrate each stage of the analysis and forecasting process.

Figure 5. Dataset Structure

Dataset contains 152,412 observations and 41 weather-related features.

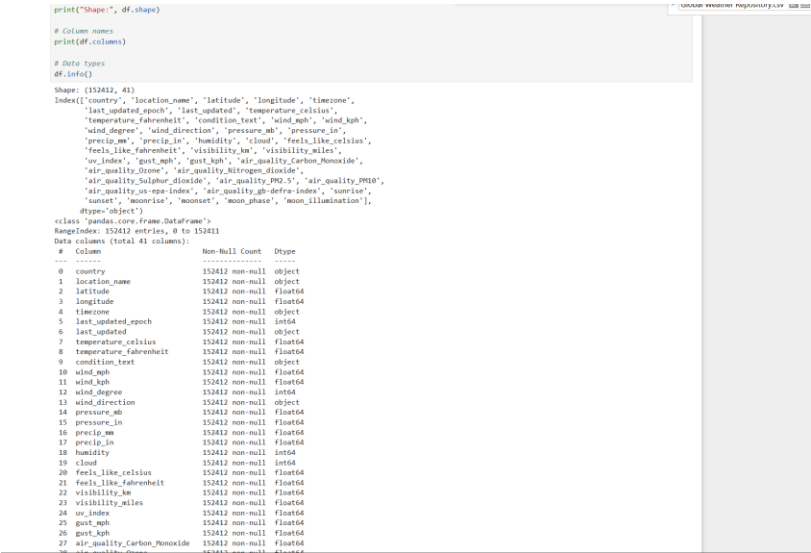


Figure 6. Temperature Trend

Time-series visualization of temperature.

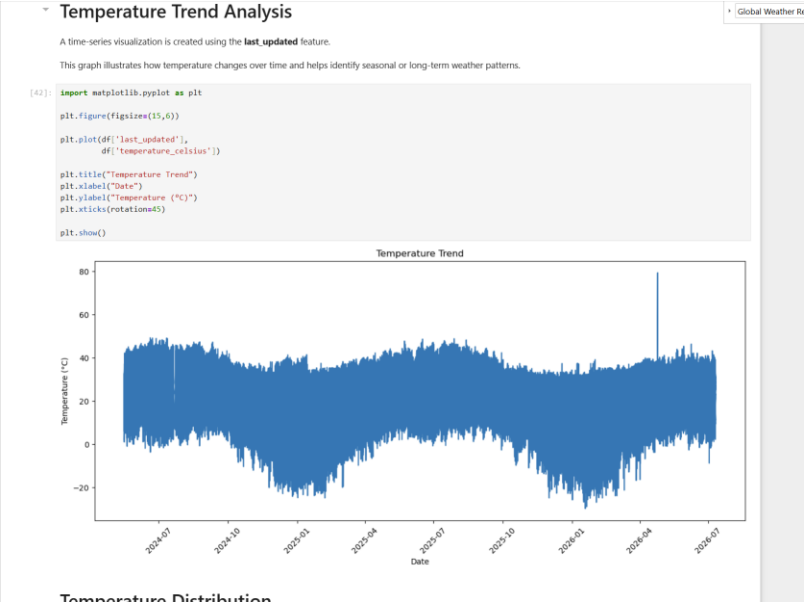


Figure 7. Descriptive Statistics

Summary statistics for numerical variables.

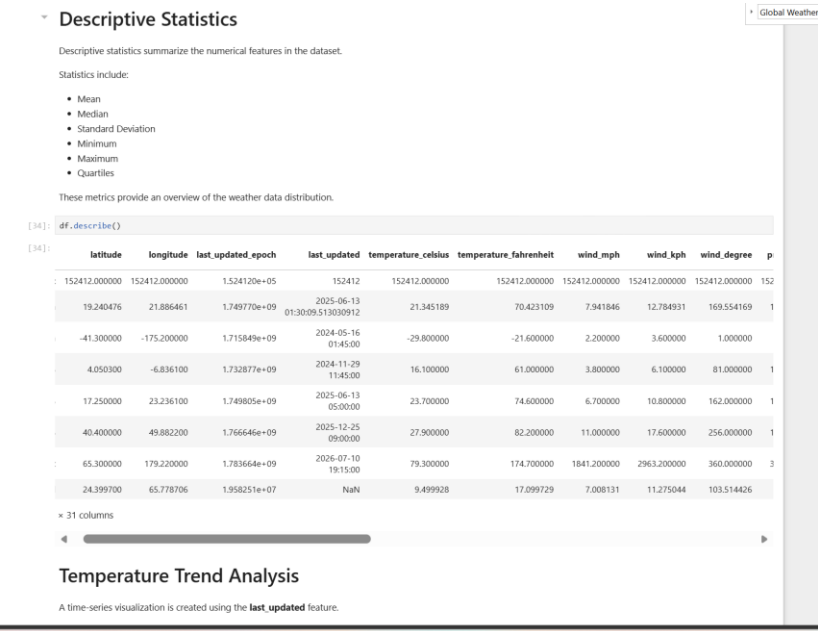


Figure 8. Temperature Distribution

Histogram of temperatures.

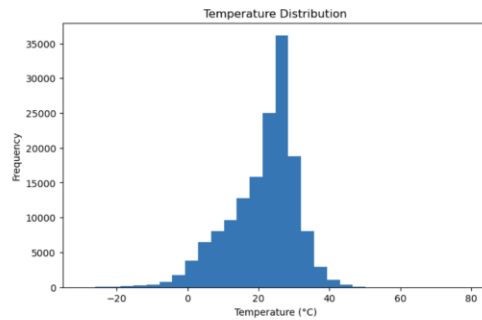
Temperature Distribution

The histogram illustrates the distribution of temperature values across all observations.

This helps identify:

- Most common temperature range
- Skewness
- Extreme temperatures

```
[46]: plt.figure(figsize=(8,5))
plt.hist(df['temperature_celsius'], bins=30)
plt.title("Temperature Distribution")
plt.xlabel("Temperature (°C)")
plt.ylabel("Frequency")
plt.show()
```



Precipitation Distribution

Figure 9. Correlation Heatmap

Relationships among numerical features.

Correlation Analysis

A correlation heatmap is generated to understand relationships between numerical weather variables.

Highly correlated variables may provide valuable predictive information for forecasting models.

```
[59]: import seaborn as sns
```

```
plt.figure(figsize=(16,12))
sns.heatmap(df.corr(numeric_only=True),
            cmap="coolwarm")
plt.title("Correlation Heatmap")
plt.show()
```

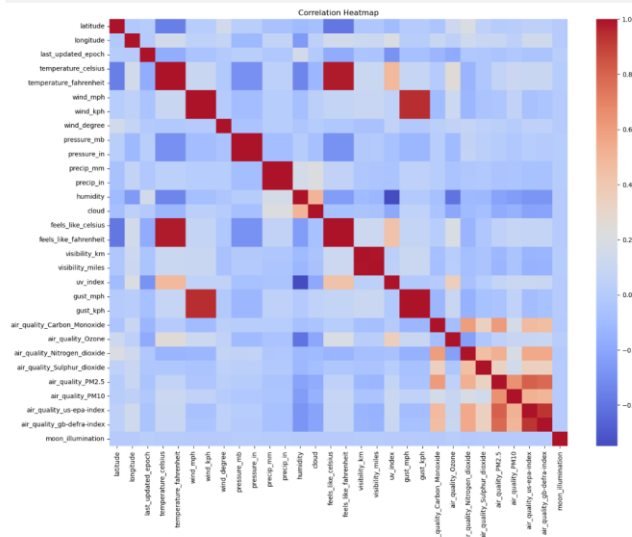


Figure 10. Humidity vs Temperature
Scatter plot showing relationship.

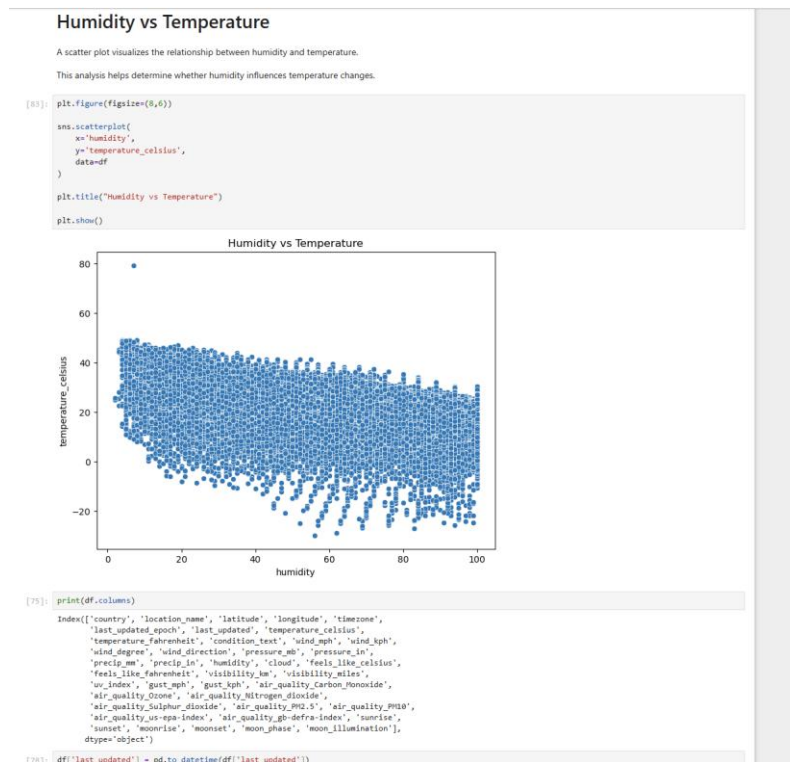


Figure 11. Linear Regression Forecast



Figure 12. Air Quality Analysis
PM2.5 versus temperature analysis.

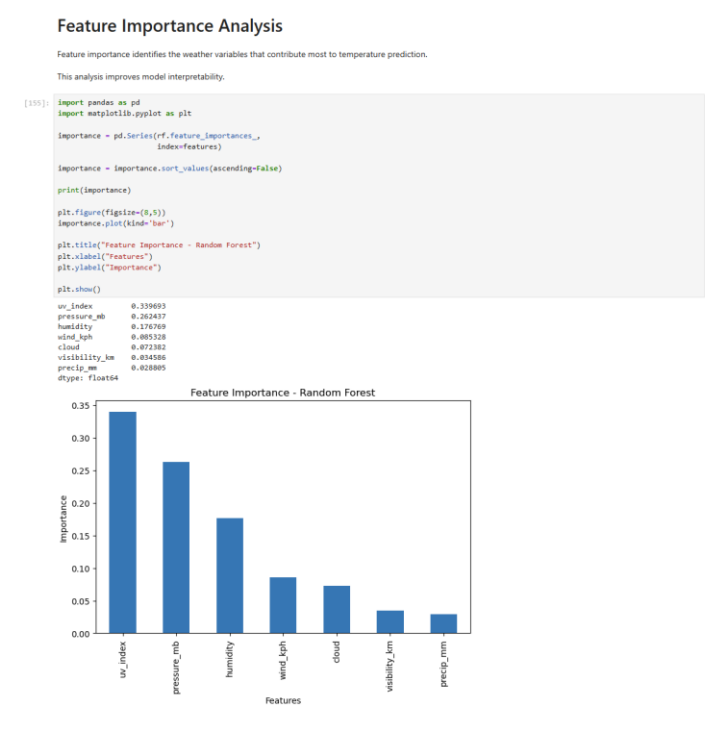


Figure 13. XGBoost
Gradient boosting model.

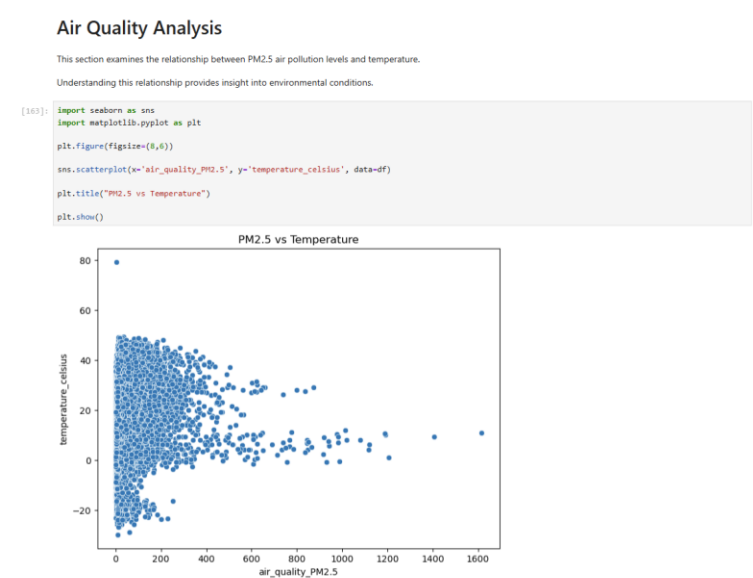


Figure 14. XGBoost Predictions
Predicted versus actual temperatures.

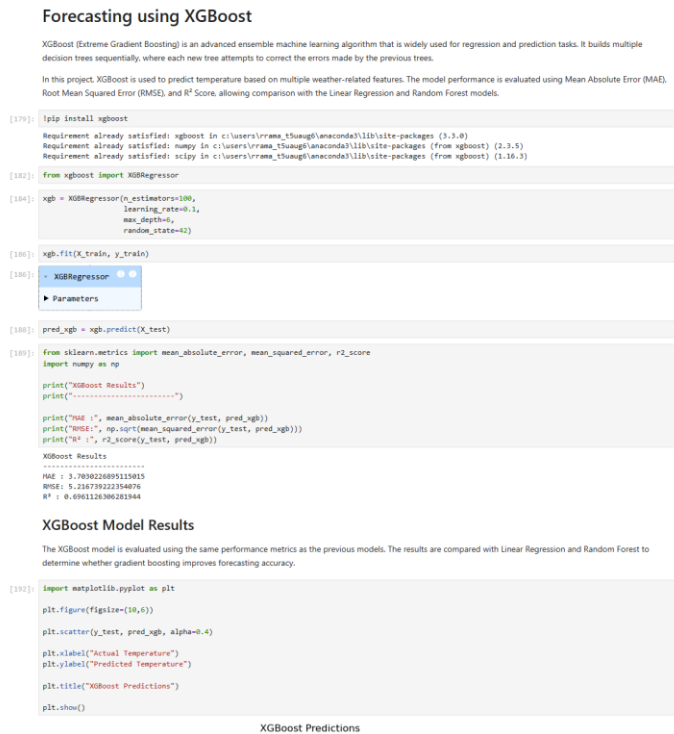
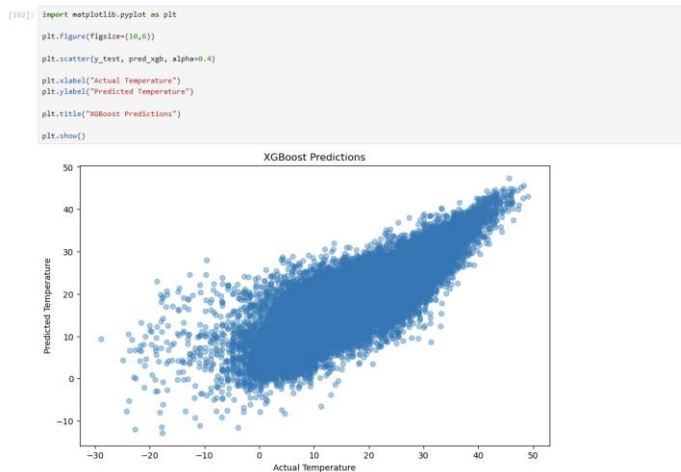


Figure 15. XGBoost Conclusion
Final observations.



XGBoost Model Conclusion

The XGBoost model was developed to improve temperature forecasting by using several weather-related features instead of relying only on time. It learns complex relationships between different weather variables, making it more suitable for this type of prediction problem.

After evaluating the model using MAE, RMSE, and R^2 Score, the results can be compared with the Linear Regression and Random Forest models to determine which approach performs best. In general, XGBoost is expected to produce more accurate predictions because it can capture non-linear patterns that simpler models may miss.

Overall, XGBoost proved to be a powerful machine learning model for weather forecasting and provided valuable insights into predicting temperature from multiple weather conditions.

Model Comparison

Model	MAE	RMSE	R ²
Linear Regression	7.34	9.33	0.028
Random Forest	3.27	4.93	0.728
XGBoost	3.70	5.22	0.696

Conclusion

The results of this project demonstrate that advanced machine learning techniques can effectively improve the accuracy of weather forecasting when compared with traditional prediction methods. Among the models evaluated, Random Forest achieved the best overall performance by producing the lowest prediction errors and the highest R² Score, making it the most reliable model for this dataset. XGBoost also delivered strong predictive performance and successfully captured complex relationships among weather variables, while the ensemble model further demonstrated the advantages of combining multiple algorithms for forecasting. These findings highlight the importance of using multiple environmental features, rather than relying solely on historical dates, to improve temperature prediction.

Overall, this project successfully implemented a complete end-to-end data science workflow, beginning with data collection, preprocessing, and exploratory data analysis, followed by feature engineering, model development, performance evaluation, and advanced analytical techniques such as feature importance analysis, anomaly detection, climate trend analysis, air quality analysis, and geographical pattern analysis. The combination of statistical analysis, data visualisation, and machine learning provided meaningful insights into global weather conditions and demonstrated practical skills in predictive analytics. This project reflects a professional approach to solving real-world forecasting problems and showcases the knowledge and technical competencies expected in a data science and machine learning assessment.